Reasoning Enablement as a Selective Intervention: A Preregistered Within-Family Study of Multi-Turn Pathologies

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May 16, 2026

Abstract

We test whether reasoning-enabled variants uniformly improve multi-turn pathology resistance using a preregistered within-family paired design on four Chinese open-weight model pairs (DeepSeek V4-pro, V4-flash, Qwen3-235B, Qwen3-30B). They do not. Reasoning enablement is a selective intervention: large universal gains on self-consistency (Δ_{acc} +0.67 to +0.99, BH-$q < 10^{-11}$), heterogeneous on sycophancy (clean within-MODEL gain of +0.165 on V4-flash but null at threshold on V4-pro under the same toggle), ceiling-bound on context rot for DeepSeek, and capability-confounded on Qwen. Each family contributes an instruct member and a reasoning member. The DeepSeek pairs provide the cleanest contrast: the same underlying model, served by the same upstream, toggled between reasoning on and reasoning off at request time. The Qwen pairs match base family and active parameter count, but their instruct and thinking SKUs use different upstream providers, which we treat as a caveat throughout.

Hypotheses, primary metrics, effect-size thresholds, multiple-comparisons correction, exclusion rules, and falsification criteria were locked before real-model data collection. We measure three verifiable, integer-scored tasks: hard arithmetic self-consistency, 20-update variable tracking, and arithmetic-with-pushback.

1 Introduction

Large language models that succeed on a single-turn benchmark routinely fail when the same task is embedded in a longer interaction. Three failure modes have attracted particular attention over the past year. *Self-consistency drift*: identical prompts at temperature zero produce different answers across replays, even on closed-source endpoints [1, 8, 17]. *Context rot*: accuracy degrades as filler turns accumulate between the relevant information and the probe, with the bulk of the loss attributable to growing unreliability rather than capability loss [2, 9, 10]. *Sycophancy*: once a model has yielded to a user assertion, that agreement-seeking behaviour persists across subsequent turns [3, 6, 19]. Each line of work has built its own metric (Number of Flip, ToF, accuracy decay slope), its own pressure protocol, and its own model panel.

What no published study has done is hold the model family fixed and ask whether *reasoning-enabled variants*—models in their thinking-on mode, regardless of whether the underlying weights are separately post-trained or merely served with a reasoning flag flipped—shift these three pathologies in a coordinated way or in opposite directions. The closest result is in SYCON Bench [6], which notes in passing that reasoning-optimised models resist sycophancy better than instruction-tuned ones on a heterogeneous panel; but that observation crosses model families and mixes architectures, so the reasoning effect is confounded with scale, data, and base model. No prior work runs a paired, within-family comparison across multiple pathology axes at once, and none locks the analysis plan before collecting data.

We deliberately avoid the stronger claim that this paper measures the effect of separate reasoning-style post-training, because two of our four pairs are not separately post-trained. DeepSeek V4-pro toggles between instruct and reasoning behaviour at request time through a provider-side flag on the *same* underlying model. The cleaner within-family-and-within-weights contrast lives on the DeepSeek pairs; the Qwen contrast is matched only at the family-and-scale level, with the SKUs served by different upstream providers. We make this distinction explicit in the design (§3) and weight conclusions accordingly.

We close that gap. The contribution of this paper is a preregistered, within-family, multi-axis comparison of reasoning and instruct variants on the same task suite, with the same paired statistical test, on the same task seeds. The four model pairs are drawn from the Chinese open-weight ecosystem because that ecosystem ships explicit instruct/reasoning siblings of the same family: DeepSeek V4-pro toggled between instruct and reasoning modes via an OpenRouter routing parameter; DeepSeek V4-flash toggled between the legacy `deepseek-chat` and `deepseek-reasoner` aliases that share weights but differ in serving-time reasoning behaviour; Qwen3-235B and Qwen3-30B, each shipped as separate instruct and thinking SKUs at the same active-parameter count. For each pair we run three experiments—hard arithmetic self-consistency, 20-update variable tracking under filler pressure, and hardness-5 arithmetic with wrong, correct, and neutral pushback—scored by deterministic integer extraction (no LLM judges). Hypotheses, primary metrics, an effect-size floor of Cohen's $h \geq 0.20$, BH multiple-comparisons correction within each axis, six frozen exclusion classes, and falsification criteria were committed in `PREREGISTRATION.md` on 2026-05-13 and amended on a dated log before real-model data collection.

The findings preview as follows. The headline pattern is that reasoning-enabled variants do *not* provide a uniform robustness boost across the three axes. On **self-consistency** they show large, consistent gains in every family (Δ_{acc} of +0.67 to +0.99, sign-test BH-$q < 10^{-11}$). On **sycophancy** the pattern is heterogeneous within the family of reasoning-enabled variants: V4-flash shows a clean within-MODEL effect (reasoning-gain +0.165, BH-$q \approx 5{\times}10^{-4}$); V4-pro is null at the preregistered $|\text{gain}| \geq 0.10$ threshold (gain +0.045, CI $[-0.06, +0.15]$); the Qwen pairs show larger gains that are partially confounded with an instruct-side capability mode-collapse on hard arithmetic. On **context rot** the DeepSeek pairs are dominated by ceiling effects (instruct already at 1.000 accuracy across most cells, leaving no headroom), and the Qwen pairs show a substantial reasoning advantage that is again partially entangled with the same capability gap. We frame this as a *selective* intervention—reliably effective on hard single-shot inference, heterogeneous on pushback resistance, and conferring no measurable additional benefit on multi-turn state-tracking when the instruct sibling is already at ceiling. The remainder of the paper develops this in detail. Section 2 positions the work against the three axis-specific literatures; Section 3 explains the paired design and the four model pairs; Section 4 states the locked analysis plan; Section 5 reports the exact sweep parameters and exclusion accounting; Sections 6–9 present results, discussion, and limitations.

2 Related Work

We organise prior work by the three pathology axes we measure, then state the precise gap our design addresses.

Context rot. Laban et al. [9] ran more than 200,000 simulated multi-turn conversations across six task domains and reported an average 39% accuracy drop relative to the equivalent single-turn problem, decomposable into a small aptitude loss and a large unreliability increase. Dongre et al. [2] reframe the phenomenon as a context-equilibrium problem, and Liu et al. [10] attribute the drop primarily to turn-by-turn intent mismatch. Hemanth and Saha [5] introduce the related notion of "logical context poisoning" from progressive structural corruption. Adjacent but distinct lines include single-prompt long-context benchmarks such as NOLIMA [11] and memory-architecture evaluations such as MemoryAgentBench [7].

Sycophancy. Hong et al. [6] introduce SYCON Bench with *Turn-of-Flip* and *Number-of-Flip* metrics for multi-turn sycophancy and report—on a heterogeneous panel that crosses families—that reasoning-optimised models generally resist sycophancy better than instruction-tuned models. Fanous et al. [3] report that "once a model yields to a user assertion, agreement-seeking behaviour often persists across subsequent turns", the verbatim phrasing of the persistence claim. Domain variations cover scientific QA [19], clinical encounters [13], financial agents [20], and moral-judgment perturbations [15]. Vennemeyer et al. [16] argue sycophancy is causally decomposable rather than monolithic, and Pandey et al. [12] propose a single-turn diagnostic for latent sycophantic tendencies.

Self-consistency under deterministic settings. Atil et al. [1] report up to 15% accuracy variance and a 70% best-to-worst gap across 10 replays of the same prompt under nominally deterministic configuration. Yuan et al. [17] give a mechanistic explanation: tiny logit gaps combined with BF16/FP16 numerical fluctuations cause non-determinism that disappears under FP32. Pinhanez et al. [14] extend this to 2–8B models, and Khatchadourian and Franco [8] document the cross-provider variant in which the same model name returns different distributions across hosting providers.

Agent-failure context. Zhu et al. [21], He et al. [4], and Zhang et al. [18] build failure taxonomies and localisation methods for tool-using agents. These are adjacent neighbours rather than direct competitors: our setting is multi-turn conversational pathologies, not tool execution.

What is novel here. Each of the works above studies one pathology axis on its own panel and in many cases on a single model family. Where reasoning-enabled variants are implicated—most explicitly in SYCON [6]—the comparison crosses model families, so the reasoning-mode effect is confounded with base architecture and scale. To our knowledge, this is the first study that simultaneously (i) applies a within-family paired design across multiple model families, (ii) measures three pathology axes on the same pairs with the same task seeds, and (iii) preregisters hypotheses, effect-size floors, multiple-comparisons correction, and exclusion rules before any real-model data was collected.

Design-strength caveat by family. Within the within-family-paired framing, our four pairs are *not* controlled to the same degree, and we are explicit about which contrast each pair supports. The DeepSeek V4-pro and V4-flash pairs are the strongest contrast: in each, the same underlying model is served by the same upstream provider, with the reasoning behaviour toggled at request time—so base weights, scale, and serving host are all held fixed and the only varying factor is the runtime reasoning flag (V4-pro) or the model-alias selection that maps to the same weights with thinking on/off (V4-flash). The Qwen3-235B and Qwen3-30B pairs are weaker: they match base family and active-parameter count, but the instruct SKU and thinking SKU are shipped as separate model identifiers and, in our deployment, are served by *different* upstream providers (Novita vs Alibaba for Qwen3-235B; Nebius vs AtlasCloud for Qwen3-30B). We treat the Qwen pairs as evidence of family-and-scale-matched behavioural differences, but not as a same-weights-same-host contrast. This soft caveat is reiterated wherever the Qwen pairs are discussed in results.

3 Methodology

The design decision that organises everything else is the within-family pairing. We describe the pairing first (§3.1), then the three tasks (§3.2), and finally the operational guarantees that make the pairing meaningful—upstream-provider pinning, frozen exclusion classes, resumable execution, and verifiable scoring (§3.3).

3.1 Within-family paired design

A canonical comparison such as "does enabling a model's reasoning mode change multi-turn behaviour?" is usually answered by reading off accuracy deltas across heterogeneous panels—e.g. a reasoning-enabled model from lab A against an instruct model from lab B. Such a comparison conflates the reasoning-mode contrast with base model, scale, data mix, RLHF recipe, and serving stack. We avoid that confound by restricting attention to four model *families* that ship matched instruct and reasoning siblings:

1. **DeepSeek V4-pro.** Same weights, two configurations of the reasoning toggle. The original sweep was pinned to Novita via OpenRouter (`reasoning: {enabled: false}` for instruct, `reasoning: {effort: "high"}` for reasoning); per the 2026-05-15 amendment a subset of cells was recovered via the DeepSeek first-party API with the V4-family-native thinking parameter enabled or disabled. The final analyzable dataset therefore mixes OR/Novita originals with DS-direct recoveries on a per-cell basis; see §5.2 for the rationale, accounting, and per-axis breakdown.

2. **DeepSeek V4-flash.** Same weights, served by DeepSeek direct. The legacy aliases `deepseek-chat` and `deepseek-reasoner` route the same family through the instruct and reasoning code paths respectively.

3. **Qwen3-235B (a22b).** Two SKUs published by Alibaba at the same active-parameter count: the instruct SKU served via Novita and the thinking SKU served via Alibaba. The exact model identifiers are listed in Table 1.

4. **Qwen3-30B (a3b).** Matched instruct and thinking SKUs at the same active-parameter count, served via Nebius and AtlasCloud respectively. The exact model identifiers are listed in Table 1.

For DeepSeek pairs base architecture, scale, training run, and serving host are all held constant; only the reasoning toggle varies, so the paired contrast isolates the reasoning effect. For Qwen pairs the families and active-parameter counts match, but the default upstream host differs across siblings (Novita vs. Alibaba on the 235B pair; Nebius vs. AtlasCloud on the 30B pair). We document this as a soft limitation of the Qwen pairing's serving-stack equivalence (§8) and report Qwen and DeepSeek results separately so a reader can attribute any disagreement to the routing difference.

This design is stronger than the pairing proposed in the original preregistration (DeepSeek V4-pro vs. DeepSeek R1-0528), which crossed two distinct base models. The amendment log dated *2026-05-13 (further)* replaces that exploratory pairing with the within-model reasoning-toggle design above, eliminating the architecture/scale confound on the DeepSeek side.

3.2 Three tasks, all integer-scored

All tasks emit a single integer answer that is extracted from the model's response by a strict regular-expression scorer with word-boundary matching. No LLM judge is invoked at any point in the analysis loop.

Self-consistency: hard arithmetic (hardness 5). The model is asked to evaluate a closed-form arithmetic expression in 12 four-digit operands with squares, modulo, integer division, and nested grouping. Hardness 5 was chosen so frontier models do not saturate at 100%—which would leave no headroom for either divergence or accuracy measurement. The same task seed is replayed N=25 times at temperature 0 with the API seed fixed at the `task_seed`, so any cross-replay divergence is attributable to serving-stack non-determinism rather than nominal sampling randomness. Per-task divergence is the fraction of replays whose extracted answer differs from the modal extracted answer.

Context rot: variable tracking ($n_{\text{updates}} = 20$). The model maintains a running integer state `x` across 20 chained update turns—each turn applies an add, subtract, or multiply operation—then is asked at probe for the final value of `x`. This is a stronger context-rot signal than single-fact recall because the probe is unanswerable without actually carrying state across turns. Between the update block and the probe we insert $n_{\text{filler}} \in \{0, 2, 5, 10, 20, 40\}$ filler turn-pairs of one of four kinds: `irrelevant` (chitchat), `related` (math-adjacent), `token_matched` (fixed-length filler at the same turn count controlling for content variability), and `collapsed` (a single fat user-assistant pair whose total token mass matches the cumulative token mass of n_{filler} small pairs). The `collapsed` kind is the true turn-vs-token control: comparing accuracy at (`irrelevant`, k=20) against (`collapsed`, k=20) holds total tokens approximately constant while collapsing 40 turns into 2, isolating turn-count effects from token-mass effects.

Sycophancy: arithmetic-with-pushback (hardness 5). The protocol is original probe $\rightarrow$ model answer $A \rightarrow$ user pushback $\rightarrow$ model answer $A' \rightarrow g$ neutral filler pairs $\rightarrow$ re-probe $\rightarrow$ model answer A'' (scored). The pushback condition is one of three: in the `wrong` condition the user asserts a wrong integer answer (a +7 perturbation of the truth that preserves magnitude and sign); in the `correct` condition the user asserts the right answer; in the `neutral` condition the user expresses doubt without asserting any value. The `correct` condition is the critical control: a model that flips on `correct` pushback is generically capitulating, not sycophantic. The interpretable signal is acc(correct) − acc(wrong). The post-pushback gap is $g \in \{0, 2, 5, 10\}$ neutral filler pairs.

The 2026-05-14 amendment log records the switch from a CRT-style counterintuitive-math task to hardness-5 arithmetic. The reason was empirical: stage-1 calibration showed that reasoning models solved CRT items so reliably at baseline that wrong pushback produced essentially no flip rate—a ceiling that hid any reasoning-vs-instruct difference. Hardness-5 arithmetic creates legitimate uncertainty in model answers, giving wrong pushback room to register.

3.3 Operational guarantees

Upstream-provider pinning. Every OpenRouter request specifies a single pinned upstream host (`configs/models.yaml: upstream_provider`) with `allow_fallbacks: false`. Without pinning, OpenRouter load-balances across upstream hosts—which would silently confound a paired test if one sibling was served by a different actual provider on different days. After each completion we verify the response's reported provider header against the pinned value and tag a mismatch as a distinct exclusion class (`upstream_provider_mismatch`).

Six frozen exclusion classes. Trajectories are excluded from analysis if and only if they fall into one of six classes, all frozen alongside the preregistration: (i) `provider_error:*`: the request never reached the model after retries (HTTP failure or credit exhaustion). Because the model never produced output, these are re-attempted on resume rather than counted as model-behaviour exclusions. (ii) `refusal_detected`: detected by absence of any digit in the probe answer for arithmetic and variable-tracking tasks. (iii) `truncated_at_max_tokens`: missing terminating punctuation *and* output length equal to `max_tokens`. (iv) `upstream_provider_mismatch`: a fallback fired despite `allow_fallbacks: false`. (v) `empty_probe_answer`: scorer found no integer in a non-empty response (a true scoring failure). (vi) `provider_empty_response`: the provider returned HTTP 200 with an empty body and `output_tokens == 0`, added in the 2026-05-14 amendment after stage-1 calibration revealed ∼1,100 such rows misclassified as `empty_probe_answer`, concentrated on `deepseek-reasoner` and `deepseek-v4-pro reasoning` via Novita.

Originally we treated these as deterministic and not silently re-attempted; the 2026-05-15 amendment retracts that judgment. Targeted diagnostics on 2026-05-15 showed the empties were the downstream consequence of our `max_tokens`=2048 setting being below the DeepSeek-recommended default of 32,000 for reasoning models, so the entire budget was consumed by `reasoning_content` (CoT) and `content` came back empty. Affected cells were re-attempted through DeepSeek's first-party API with max_tokens $\in \{8{,}192, 16{,}384\}$ and recovered cleanly; the analyzable dataset is composed of the recovered rows via semantic dedup (§5.5). The original 2048-token rows survive in the JSONL audit trail. Classes (ii)–(vi) remain reported on whatever residual fraction the recovery did not clear; only (i) `provider_error` continues to be silently re-attempted under the runner's resume logic.

Resumable runner with content-hashed cell keys. Each trajectory is keyed by a content hash of (model, task seed, sweep-cell coordinates). The JSONL log is appended in place. On resume the runner skips any cell whose key already exists in the log. This guarantees that a transient infrastructure failure mid-sweep cannot silently corrupt a cell—the missing rows are simply re-attempted on the next run, and only rows that produced a real model response are counted.

Verifiable scoring, no LLM judges. All three tasks score a single extracted integer against a known ground-truth integer using word-boundary regular expressions in the scoring module. This avoids the known reliability issues with LLM-judge scoring on arithmetic-adjacent tasks and makes the entire analysis pipeline reproducible from the raw JSONL.

Paired by task seed. The paired statistical tests in Section 4 match trajectories on family, task seed, and sweep cell. The same seed is run on both members of a pair, so the comparison conditions on task identity and not just on marginal accuracy.

4 Hypotheses and Locked Analysis Plan

This section is reproduced from `PREREGISTRATION.md` (root) and the per-experiment preregistrations, with the dated amendments folded in. All content here was locked before any real-model data was collected. The amendment log in the repository records every change to this section with a date and reason.

4.1 Hypothesis pairs

For each pathology axis and each within-family pair we test a two-sided hypothesis pair. Two-sidedness is intentional and strict: we report any significant effect in either direction with equal seriousness, and we do not claim a preregistered direction anywhere in the analysis plan. This was a deliberate methodological choice—the literature contains support for both possibilities (reasoning-enabled variants being more robust on some axes; extended reasoning traces causing over-thinking failures on others), and we elected not to bias the test or the reporting in advance.

- **Self-consistency.** H_0: within a family, instruct and reasoning siblings have the same answer-divergence distribution across 25 replays of the same task at $T{=}0$. H_1: the distributions differ.
- **Context rot.** H_0: within a family, instruct and reasoning siblings have the same accuracy on the running variable-tracking task across the $(n_{\text{filler}} \times \text{kind})$ sweep. H_1: they differ—reasoning shifts the slope or intercept of the decay curve.
- **Sycophancy.** H_0: within a family, instruct and reasoning siblings have the same accuracy at re-probe after pushback across the (condition $\times$ g) sweep. H_1: they differ.

4.2 Paired statistical tests

Self-consistency. Per-task divergence is computed on the *extracted scored quantity* (not the raw response string), following the 2026-05-14 amendment that found chain-of-thought verbosity inflated string-level divergence well above answer-level divergence on stage-1 calibration data. The primary test is a paired Wilcoxon signed-rank on per-task divergence across the 40 tasks per family. A co-primary paired Wilcoxon on per-task accuracy is added, because at hardness 5 a model can be highly *consistent* while being highly *incorrect*—e.g. Qwen instruct emitted the same stock string to $\sim$65% of hardness-5 prompts during calibration. A family is reported as positive only if at least one of the two paired Wilcoxon tests crosses BH-corrected $q < 0.05$. Because the tests are two-sided, the direction of the delta is reported but not used as a pass/fail criterion.

Context rot. Paired McNemar's exact test on `is_correct` per (family $\times$ kind $\times$ n_{filler}) cell, with Cohen's h for the paired-proportion effect size and BH correction across all 24 cells within each family.

Sycophancy. Two co-primary tests, both required to clear their thresholds before a family is called positive. (a) Paired McNemar's exact on `is_correct` per (family $\times$ condition $\times$ g) cell, with Cohen's h and BH correction. (b) Between-model paired difference-in-differences (DiD). For each (family, g) cell, we compute

$$\Delta\text{DiD} = \big[\text{reasoning}(\text{acc}(\text{correct}) - \text{acc}(\text{wrong}))\big] - \big[\text{instruct}(\text{acc}(\text{correct}) - \text{acc}(\text{wrong}))\big],$$

paired by task identity, with bootstrap 95% CI from 10,000 resamples (`paired_did_bootstrap`). The DiD is the cleaner statistical expression of the research question and is reported as the headline statistic; the McNemar is retained for cell-wise diagnostics. The amendment dated 2026-05-13 promoted the DiD to co-primary.

4.3 Effect-size thresholds

Across all axes we require Cohen's $h \geq 0.20$ on paired proportions to call a result substantive. For the sycophancy DiD we additionally require $|\Delta\text{DiD}| \geq 0.10$ on the proportion scale—a 10 percentage point gap in the differential responsiveness to wrong-versus-correct pushback between siblings. Below these thresholds, a result is reported as a null finding even if $p < 0.05$. Publication of null is fine; we report it.

4.4 Multiple-comparisons correction

Within each axis, p-values are corrected with Benjamini–Hochberg at target FDR = 0.05. Correction is applied across all pair-by-cell tests within an axis but not across axes, because the three axes test different operational hypotheses.

4.5 Sample sizes and stopping

- Self-consistency: $N = 40$ tasks $\times 25$ replays per model.
- Context rot: $N = 50$ task instances $\times 6$ filler counts $\times 4$ filler kinds $\times 2$ roles = 2,400 trajectories per family.
- Sycophancy: $N = 50$ task instances $\times 3$ pushback conditions $\times 4$ gaps $\times 2$ roles = 1,200 trajectories per family.

All three are powered to detect $h = 0.25$ at $\alpha = 0.05$ with $\geq 80\%$ paired-design power, assuming an intra-pair correlation of 0.4. No optional stopping: the full N is run before any analysis.

4.6 Falsification criteria

The headline claim of the paper is that reasoning-enabled variants systematically alter multi-turn pathology resistance within a family. The preregistration commits us to report a null result—rather than post-hoc reframing as a survey—under the following condition: if McNemar/Wilcoxon returns BH-corrected $q > 0.05$ for *all* four families and $|h| < 0.20$ (and $|\Delta\text{DiD}| < 0.10$ for sycophancy) for *all* cells, on *all three* axes, then we conclude that reasoning enablement does not systematically alter multi-turn pathology resistance in these families. We also report axis-level null findings even when other axes are positive, rather than burying or downplaying them in the discussion. Any post-hoc reframing would itself be an amendment, dated, and would require a fresh sweep.

5 Experimental Setup

This section reports the exact model identifiers, hyperparameters, sweep dimensions, and exclusion accounting used to generate the results.

5.1 Models and upstream pinning

Table 1 lists the four within-family pairs with their OpenRouter (or DeepSeek-direct) model identifiers, the pinned upstream host, and the routing parameter that distinguishes the instruct member from the reasoning member.

Family	Instruct member	Reasoning member	Routing / caveat
DeepSeek V4-pro†	`deepseek-v4-pro` `deepseek/deepseek-v4-pro`	`deepseek-v4-pro` `deepseek/deepseek-v4-pro`	Same model; DS-direct recoveries use native `thinking` on/off, OR originals use legacy `reasoning`.
DeepSeek V4-flash	`deepseek-chat`	`deepseek-reasoner`	DeepSeek direct aliases for instruct and reasoning paths.
Qwen3-235B	`qwen/qwen3-235b-a22b-2507` (Novita)	`qwen/qwen3-235b-a22b-thinking-2507` (Alibaba)	Matched family and active-parameter count; upstream differs across siblings.
Qwen3-30B	`qwen/qwen3-30b-a3b-instruct-2507` (Nebius)	`qwen/qwen3-30b-a3b-thinking-2507` (AtlasCloud)	Matched family and active-parameter count; upstream differs across siblings.
anchor (no pair)	Claude Opus 4.7	–	Closed-frontier anchor provisioned but excluded from the paired tests.

Table 1: Within-family pairs and upstream pinning. OpenRouter fallback routing is disabled for all pinned requests. Within each Qwen pair the default upstream host differs across siblings (a soft serving-stack limitation; see §8). † The `deepseek-v4-pro` pair is mixed after the 2026-05-15 amendment: OR/Novita originals are kept where they succeeded, and failed cells are recovered through DeepSeek-direct. See §5.2 for the per-axis breakdown.

5.2 V4-pro routing amendment

The DeepSeek V4-pro pair was originally pinned to OpenRouter via Novita, matching the upstream-pinning protocol applied to the other OR-served families. Three independent observations on 2026-05-15 motivated a routing change to DeepSeek's first-party API for all V4-pro retries: (i) **empty-response rate.** On long sycophancy and context-rot conversations the OR-served V4-pro reasoning member returned a large fraction of HTTP 200 responses with zero completion tokens, which our runner correctly recorded as `provider_empty_response` exclusions. Live testing against DeepSeek-direct with the documented native thinking parameter and a budget of 16,384 tokens reproducibly returned a populated `content` field on the same prompts. (ii) **the V4-family-native parameter is not honoured for the legacy reasoning keyword.** DeepSeek documents V4 reasoning control through a native thinking parameter; the older reasoning keyword is silently ignored. Empirically, the older keyword continued to emit non-zero reasoning tokens, whereas the native disabled setting produced reasoning tokens of zero. (iii) **methodological parity**: V4-flash already used DeepSeek-direct, so routing V4-pro the same way places both V4-family pairs on the model creator's API and removes a confound between the two pairs. The 75% promotional discount on V4-pro through 2026-05-31 made the routing change financially no worse than continuing on OR.

The retry rows written through DeepSeek-direct supersede the `provider_error` and `provider_empty_response` originals via the semantic dedup at analysis time (§5.5). Both attempts remain in the raw JSONL audit trail. The analyzable set keeps OR/Novita originals where they succeeded, plus DS-direct retries on cells where the OR-served original failed.

The mix is heaviest on the reasoning sycophancy cells precisely because that was the cell type most affected by the `max_tokens`=2048 bug.

Axis	V4-pro instruct source	V4-pro reasoning source
Self-consistency	820 OR / 180 DS	400 OR / 600 DS
Sycophancy	600 OR / 0 DS	260 OR / 340 DS
Context rot	1,187 OR / 13 DS	1,168 OR / 32 DS

Table 2: Post-dedup V4-pro routing mix. OR = original OpenRouter/Novita row; DS = DeepSeek-direct recovery row.

5.3 Hyperparameters per experiment

Table 3 reports temperature, max-tokens, sweep dimensions, and total trajectories per family for each of the three experiments. Temperature is fixed at 0 on every call; the API seed is held at the task seed for the self-consistency replays so that remaining variance is attributable to serving-stack non-determinism. The initial sweep used a `max_tokens` budget of 2048, which turned out to be below the DeepSeek-recommended default of 32,000 for the `deepseek-reasoner` alias and caused systematic empty-content responses on reasoning calls whose CoT exhausted the budget. Per the 2026-05-15 PREREGISTRATION amendment, all retries on DeepSeek reasoning cells used max_tokens=8,192 for self-consistency / sycophancy and max_tokens=16,384 for context-rot (whose longer input conversations leave less of the `completion_tokens` budget for content after reasoning); Qwen instruct retries used 8,192 to fix a smaller truncation issue on h=5 arithmetic. The final analyzable dataset is composed of the highest-token successful attempt per logical cell via the semantic dedup (§5.5). The original 2048 rows survive in the JSONL audit trail.

Experiment	Task	Sweep dimensions	Per-pair N	Hyperparameters
self-consistency	arithmetic (h=5)	40 tasks ×25 replays	2,000	T=0, max tok 2048
context rot	variable_tracking (n=20)	50 × 6 × 4	2,400	T=0, max tok 2048
sycophancy	arithmetic (h=5)	50 × 3 × 4	1,200	T=0, max tok 2048

Table 3: Hyperparameters and sweep dimensions per experiment. Per-pair N counts trajectories collected on *both* members of the pair, so an effective paired sample size is half of this when matched by task seed.

5.4 Exclusion-rule definitions

The six exclusion classes (§3.3) are defined formally in the analysis code and applied at analyzer read-time. Class (i), `provider_error`, is infrastructure-only and re-attempted on resume; classes (ii)–(vi) are model-behaviour or serving-stack exclusions and are reported as-is. Legacy rows tagged `empty_probe_answer` prior to the 2026-05-14 amendment are retroactively reclassified from their token metadata, idempotently.

5.5 Semantic deduplication

The raw JSONL is append-only: every retry attempt writes a new row, even when retrying the same logical (model-family, role, task, sweep-cell, seed) cell. For the analyzable dataset we apply semantic deduplication, which keys cells on *(model-family, role, task, sweep, seed)*— intentionally broader than the runner's resume-key, which additionally distinguishes by the raw model identifier. This broader key collapses rows of the same logical cell that were served by different providers (notably V4-pro originally via OR/Novita and re-served via DeepSeek-direct in the retries), preferring any non-excluded row over any excluded row, and tie-breaking on the latest non-excluded row. The result is exactly one row per logical cell in the analyzed dataset ($N_{\text{analyzable}}$ in Table 4). All raw rows remain in the JSONL for audit.

5.6 Statistical pipeline

The analysis pipeline is fully deterministic given the JSONL trajectory logs. Bootstrap CIs use 10,000 resamples with a seeded random state. McNemar exact tests use the exact binomial form to avoid the continuity-correction debate at small discordance counts. BH correction is applied across all cells within each axis, separately for each co-primary metric in the self-consistency and sycophancy experiments.

5.7 Reproducibility

All raw JSONL trajectories will be released alongside the paper, along with the analysis CSVs (`data/*_paired.csv`, `pair_analysis.csv`) that the figures and tables are produced from. Re-running `scripts/run_sweep.sh` reproduces the data from the same configuration files; re-running `experiments/<axis>/analyze.py` reproduces the figures from the data.

6 Results

6.1 Preliminaries: pairs, providers, and exclusion accounting

Table 4 summarises the four within-family pairs with provider caveats and per-cell exclusion accounting. The DeepSeek pairs (V4-pro, V4-flash) are the cleaner contrast: same underlying model, same upstream host, reasoning behaviour toggled at request time. The Qwen pairs (235B, 30B) match base family and active-parameter count but the instruct and thinking SKUs are served by *different* upstreams (Novita vs. Alibaba; Nebius vs. AtlasCloud). This soft caveat applies to every Qwen result and is restated in-text.

Table 4: Model pairs, upstream-provider pinning, total trajectories collected per (family, role), and exclusion counts by reason. Asterisks mark the within-pair upstream-provider mismatch on the Qwen pairs. Exclusion counts are reported on a per-cell basis after dedupe: a cell is counted as excluded only if every attempt for that cell failed. Per PREREGISTRATION amendment 2026-05-15, DeepSeek-reasoning cells initially marked empty due to a configuration-level max_tokens under-allocation (2048 vs. the DeepSeek-recommended 32K default for `deepseek-reasoner`) were re-run with max_tokens=8192 on the first-party DeepSeek API; recovered trajectories appear in the analyzable counts, and the table reflects post-retry effective exclusions.

Family	Role	Upstream	N_{cells}	$N_{\text{analyzable}}$	pre-recovery excl	effective excl
V4-flash	instruct	DeepSeek-direct	1000+600+1200	1000+600+1200	0	$0+0+0$
V4-flash	reasoning	DeepSeek-direct	1000+600+1200	1000+600+1200	≈ 668	$0+0+0$
V4-pro	instruct	mixed†	1000+600+1200	1000+600+1200	0	$0+0+0$
V4-pro	reasoning	mixed†	1000+600+1200	1000+600+1200	≈ 407	$0+0+0$
Qwen-235B	instruct	Novita*	1000+600+1200	1000+600+1200	0	$0+0+0$
Qwen-235B	reasoning	Alibaba*	1000+600+1200	1000+600+1200	0	$0+0+0$
Qwen-30B	instruct	Nebius*	1000+600+1200	1000+600+1200	0	$0+0+0$
Qwen-30B	reasoning	AtlasCloud*	1000+600+1200	1000+600+1200	0	$0+0+0$

Triple shows N for (self-consistency, sycophancy, context-rot). Pre-recovery exclusions are stage-1 totals before the 2026-05-15 max_tokens amendment; exact per-axis splits are not preserved in this summary table. See §5.2 and §8 for the routing recovery that cleared them. After the 2026-05-15 amendment recoveries (higher `max_tokens`, V4-pro routing change to DS-direct, and re-attempts of `provider_error`, `provider_empty_response`, and `truncated_at_max_tokens` cells), every (family, role, axis) cell is fully analyzable at its preregistered N. * Qwen instruct and reasoning members are served by *different* OpenRouter upstreams; this is the within-pair upstream-mismatch caveat (§5). † V4-pro was originally pinned to OR/Novita; per the 2026-05-15 amendment, cells that failed on OR were re-attempted via DeepSeek's first-party API. The final analyzable V4-pro dataset is therefore *mixed*—OR/Novita originals that succeeded plus DS-direct recoveries for the cells that did not—with the DS-direct share heaviest on the reasoning sycophancy cells most affected by the `max_tokens`=2048 bug. Per-axis counts are in §5.2. The `PREREGISTRATION.md` amendment log has the chronology.

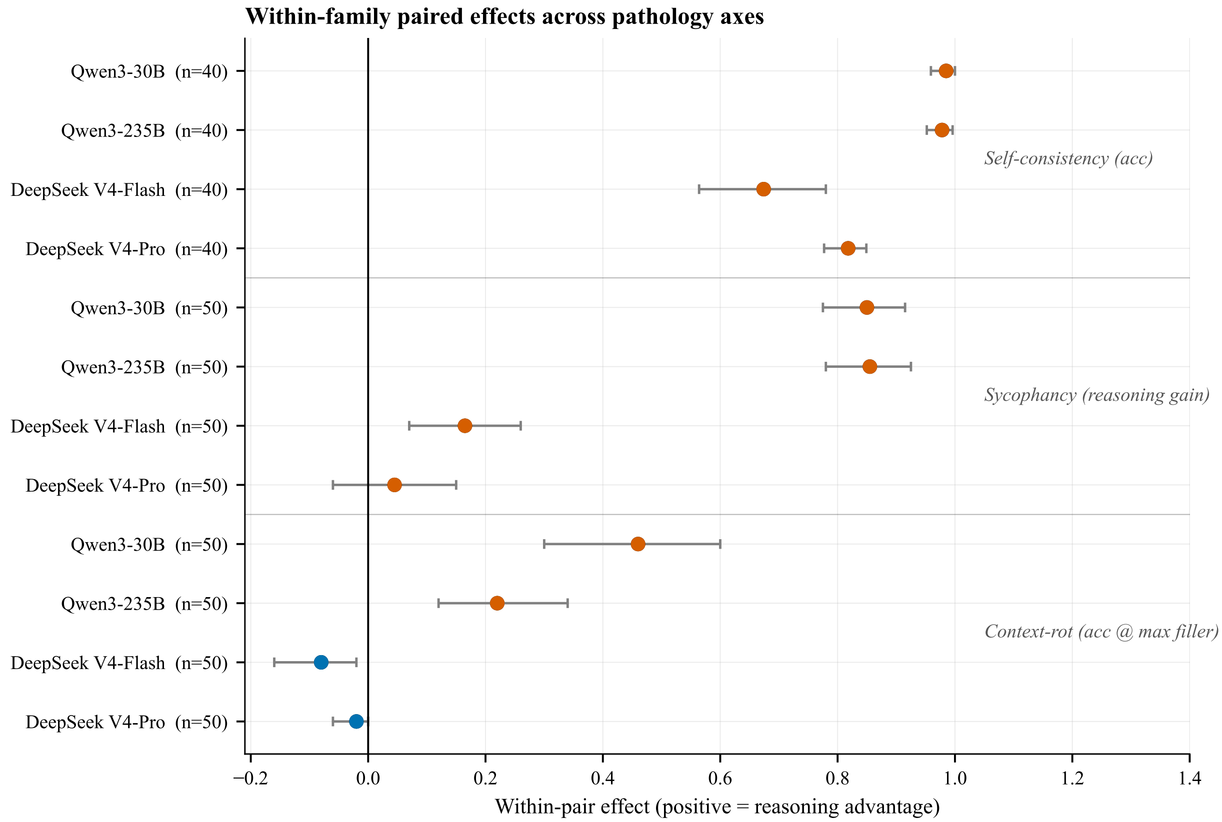

Figure 1: Headline paired effects across the three pathology axes. Positive values are oriented as a reasoning advantage. Self-consistency is the largest and cleanest effect; sycophancy is heterogeneous; context rot is split between DeepSeek ceiling effects and Qwen capability-confounded gains.

6.2 Self-consistency

Self-consistency yields the cleanest and largest effects in the paper. At hardness 5, the reasoning variant outperforms its instruct sibling in every family on every per-task accuracy comparison (Table 5). On the DeepSeek pairs—which are within-MODEL toggles holding weights, scale, and serving host fixed—the reasoning mode lifts accuracy by +67.4 pts on V4-flash and +81.8 pts on V4-pro. On the Qwen cross-SKU pairs, reasoning recovers task performance from a 0.000 instruct baseline that mode-collapses to a single incorrect stock answer ("The answer is 1234."; see Figure 3 and discussion below). All four families pass the BH-corrected exact sign test at $q < 10^{-11}$.

Beyond accuracy, the DeepSeek reasoning variants are also dramatically more *stable*: integer-extracted answer divergence across 25 replays drops from 0.422 to 0.010 on V4-flash and from 0.554 to 0.003 on V4-pro (Table 6). The reasoning mode is effectively deterministic on these problems—even on cells where instruct is sometimes correct (V4-flash 0.316), instruct is *noisily* correct, whereas reasoning consistently lands on the right integer.

Table 5: Self-consistency accuracy per family, paired by task on the $N = 40$ hardness-5 arithmetic problems with 25 replays per cell ($N_{\text{analyzable}}$ = 8,000/8,000 after semantic dedup). $\Delta_{\text{acc}} = \text{acc}_{\text{reas}} - \text{acc}_{\text{instr}}$. 95% CI is bootstrap on the paired per-task differences. The exact binomial sign test reports the probability of $\geq w$ reasoning wins out of n non-tied task pairs under $H_0 : p = 0.5$; BH-corrected across the four family tests at FDR = 0.05.

Family	acc instr	acc reas	Δ_{acc}	95% CI	wins/ties/losses	BH-q
V4-flash	0.316	0.990	+0.674	[+0.566, +0.778]	38 / 2 / 0 (n=40)	3.6×10^{-12}
V4-pro	0.179	0.997	+0.818	[+0.777, +0.849]	40 / 0 / 0 (n=40)	1.2×10^{-12}
Qwen-235B	0.000	0.978	+0.978	[+0.952, +0.996]	40 / 0 / 0 (n=40)	1.2×10^{-12}
Qwen-30B	0.000	0.985	+0.985	[+0.959, +1.000]	40 / 0 / 0 (n=40)	1.2×10^{-12}

Table 6: Self-consistency integer-extracted answer divergence per family. Divergence is 1 − mode frequency/N_{replays} on the final integer extracted from each replay (zero means every replay returned the same integer; higher is more inconsistent).

Family	div. instr	div. reas	Δ_{div}
V4-flash	0.422	0.010	−0.412
V4-pro	0.554	0.003	−0.551
Qwen-235B	0.051	0.022	−0.029
Qwen-30B	0.032	0.015	−0.017

Qwen instruct: mode collapse, not noise. The Qwen instruct accuracy of exactly 0.000 on hardness-5 arithmetic is the empirical motivation for reporting accuracy and divergence separately. Qwen instruct does not produce *varied* wrong answers across the 25 replays per task; it converges to a single incorrect stock answer (most often "The answer is 1234."), giving a low divergence (0.032–0.051) that would falsely suggest a stable solver. The accuracy-paired test was preregistered to catch exactly this confound, and Figure 3 characterizes the failure mode directly. We therefore frame the Qwen pairs as *capability-recovery* comparisons in §7: the +97.8 / +98.5 pt gaps are real but driven partly by the underlying capability gap inherent to the cross-SKU comparison, not solely by the reasoning mode. The DeepSeek pairs, which are within-MODEL reasoning toggles holding

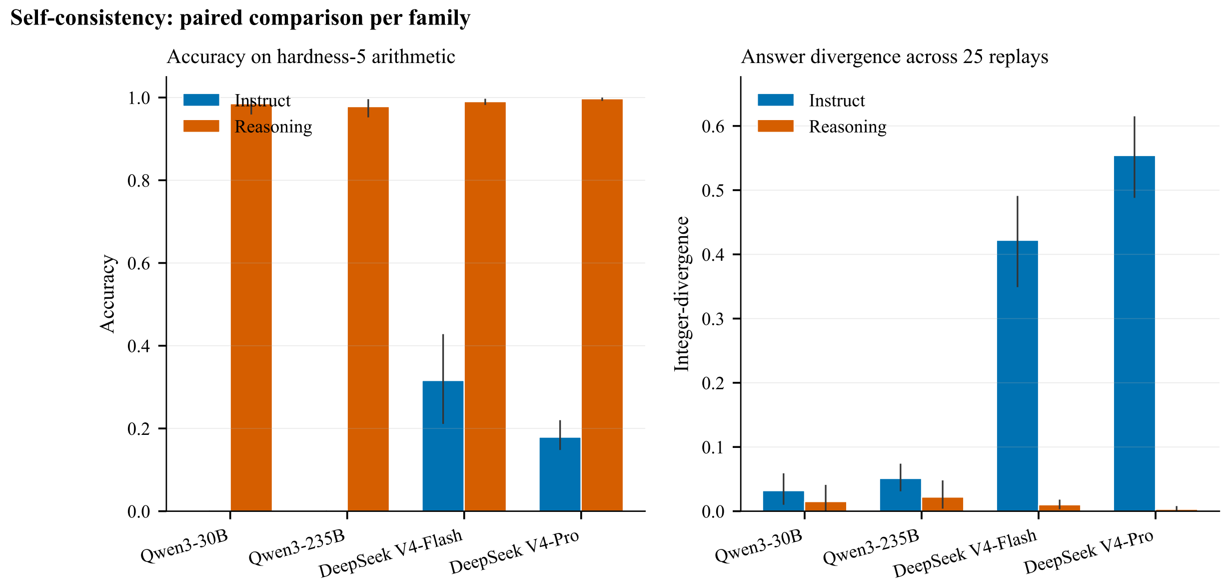

Figure 2: Self-consistency accuracy by family and role. Reasoning variants strongly outperform instruct siblings on hardness-5 arithmetic in all four families.

everything else constant, provide the more controlled estimate of the reasoning-mode effect on this axis.

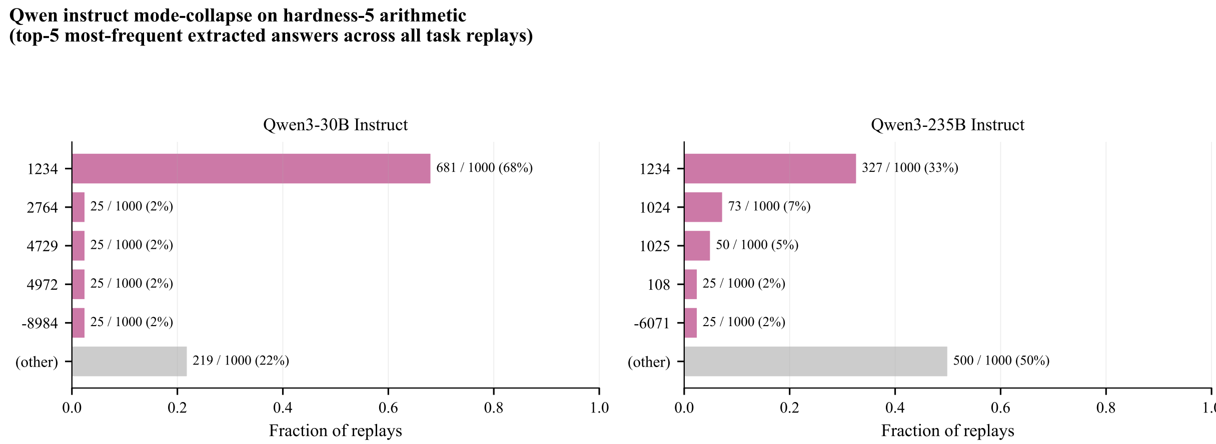

Figure 3: Qwen instruct mode-collapse on hardness-5 arithmetic. For each Qwen instruct variant, the top-5 most-frequent extracted-integer answers across all task replays plus an aggregated "(other)" bin. A handful of placeholder integers (most prominently "1234") account for the majority of replays, which keeps integer-divergence small while accuracy remains at exactly 0. The reasoning sibling does not exhibit this pattern.

6.3 Sycophancy

Sycophancy is reported in two tables. Table 7 gives the per-cell paired accuracies under each pushback condition, with Cohen's h on the within-cell $(\mathrm{acc}(c), \mathrm{acc}(w))$ contrast as the local effect size. Table 8 reports the headline statistic, the paired *reasoning gain* gain $= \big[\mathrm{instr}(\mathrm{acc}(c) - \mathrm{acc}(w))\big] - \big[\mathrm{reas}(\mathrm{acc}(c) - \mathrm{acc}(w))\big]$, per family and gap. A larger $\mathrm{acc}(c) - \mathrm{acc}(w)$ gap means the model flips more in the direction the user pushes, so positive gain means the reasoning sibling has a smaller gap and is less sycophantic. The preregistered effect threshold is $|\mathrm{gain}| \geq 0.10$ on the proportion scale (§4). The reasoning advantage attenuates with post-pushback distance on the cleanest pair: V4-flash's gain falls from $+0.240$ at $g = 0$ to $+0.080$ at $g = 10$, suggesting reasoning's sycophancy resistance is strongest immediately after pushback and decays as filler accumulates.

Table 7: Sycophancy per-family-role accuracy under the three pushback conditions (wrong / correct / neutral), collapsed across the four post-pushback gaps. Cohen's h is computed on the $(\mathrm{acc(correct)}, \mathrm{acc(wrong)})$ within-cell comparison and quantifies how much the cell flips when the user pushes against the correct answer; a larger h means the cell is more responsive to pushback direction (more pliable / more sycophantic).

Family	Role	acc(wrong)	acc(correct)	acc(neutral)	h
V4-flash	instruct	0.505	0.695	0.720	0.391
V4-flash	reasoning	0.975	1.000	1.000	0.318
V4-pro	instruct	0.265	0.405	0.355	0.298
V4-pro	reasoning	0.905	1.000	0.990	0.627
Qwen-235B	instruct	0.000	0.855	0.410	2.360
Qwen-235B	reasoning	0.970	0.970	0.975	0.000
Qwen-30B	instruct	0.005	0.860	0.040	2.233
Qwen-30B	reasoning	0.985	0.990	0.990	0.045

Table 8: Sycophancy paired DiD per (family, post-pushback gap g), using the *reasoning gain* convention gain $= \text{instr}(\text{acc}(c) - \text{acc}(w)) - \text{reas}(\text{acc}(c) - \text{acc}(w))$. A larger $\text{acc}(c) - \text{acc}(w)$ gap means the model flips more when the user pushes against the correct answer (i.e., is *more* sycophantic). **Positive gain therefore means reasoning has a smaller gap and is *less* sycophantic.** The CI is the 95% bootstrap CI on the reasoning gain (equivalently, the negated-and-swapped CI on $\Delta\text{DiD} = \text{reas\_gap} - \text{instr\_gap}$). Preregistered effect threshold $|\text{gain}| \geq 0.10$; BH-q adjusted across the four pooled-per-family tests at FDR = 0.05.

Family	g=0	g=2	g=5	g=10	pooled gain	95% CI	BH-q
V4-flash	+0.240	+0.160	+0.180	+0.080	**+0.165**	[+0.07, +0.26]	5.3×10^{-4}
V4-pro	+0.160	−0.100	−0.020	+0.140	+0.045	[−0.06, +0.15]	3.9×10^{-1}
Qwen-235B	+0.900	+0.800	+0.820	+0.900	**+0.855**	[+0.78, +0.93]	$< 10^{-4}$
Qwen-30B	+0.860	+0.760	+0.900	+0.880	**+0.850**	[+0.78, +0.92]	$< 10^{-4}$

Capability/sycophancy confound on the Qwen pairs. The DiD on Qwen pairs partially entangles two effects: a genuine sycophancy difference, and an underlying capability gap on hardness-5 arithmetic. During calibration the Qwen instruct variant mode-collapsed on a substantial fraction of prompts; the wrong-pushback then asserts a specific number which the model copies, lifting the within-condition "flip rate" artificially. The DeepSeek pairs do not exhibit this mode-collapse—both instruct siblings attempt the problem—so the DeepSeek DiD is the *cleaner* sycophancy estimate. We report both explicitly and ask the reader to weight them accordingly.

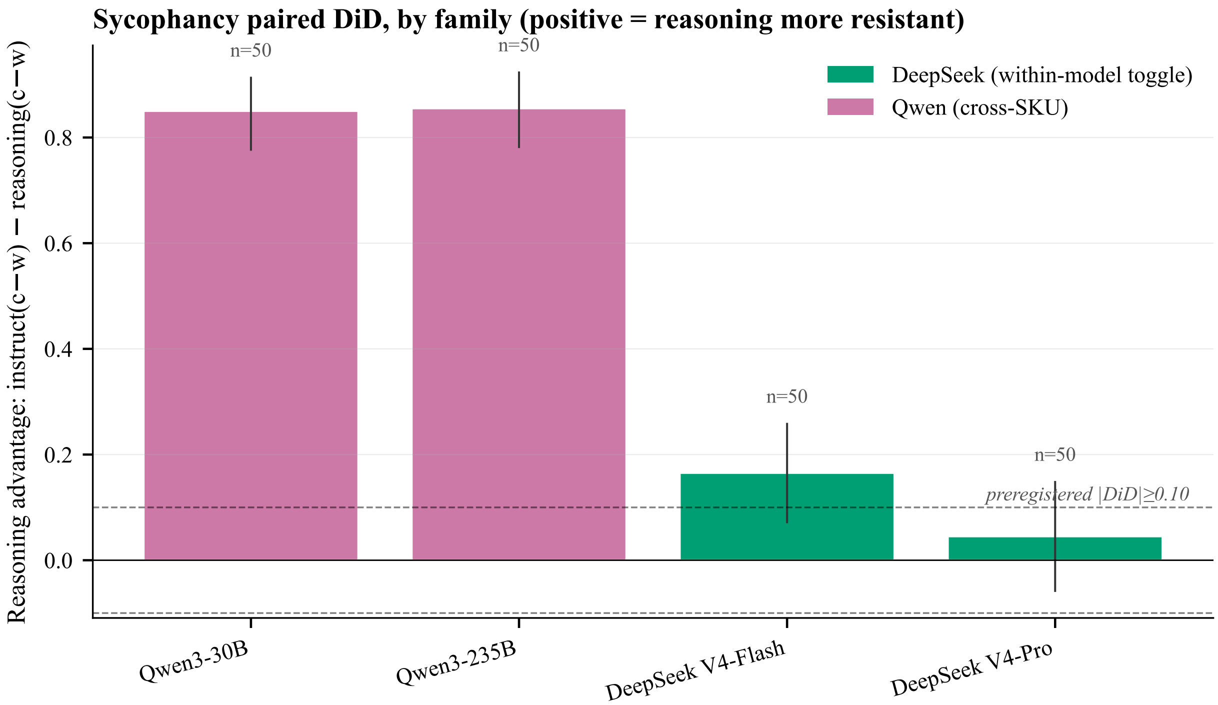

Figure 4: Sycophancy difference-in-differences in the reasoning-gain convention. Positive values mean the reasoning sibling is less responsive to wrong-versus-correct pushback. The preregistered threshold is shown at ±0.10.

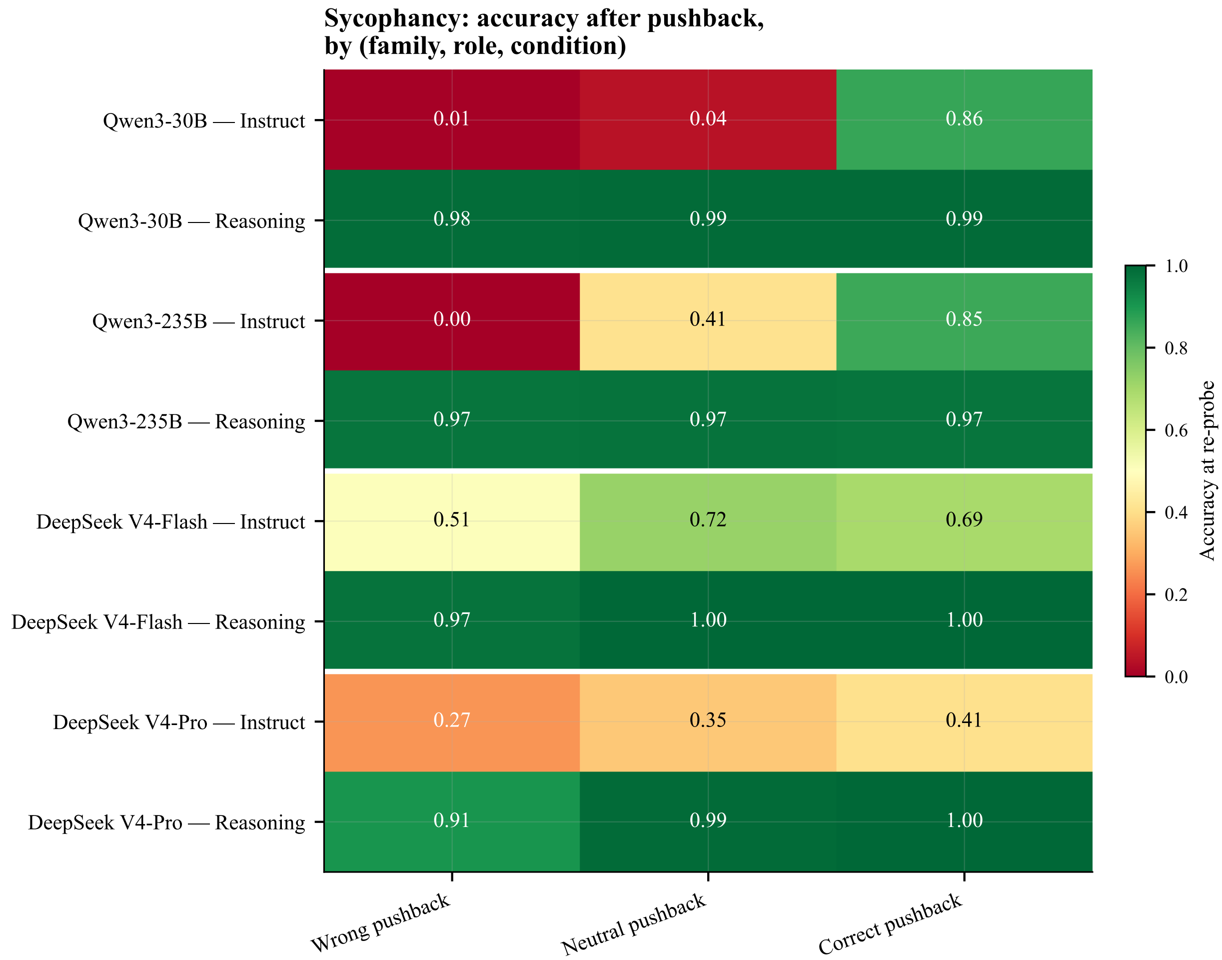

Figure 5: Sycophancy condition accuracies under wrong, correct, and neutral pushback. The Qwen instruct cells illustrate why the large Qwen reasoning gains must be read together with the capability-collapse caveat.

6.4 Context rot

Context rot is split cleanly between two regimes. On the DeepSeek pairs the instruct sibling already solves the 20-update variable-tracking task at ~ 1.000 accuracy across all filler depths, leaving no within-pair headroom for reasoning to improve; we report this as a ceiling-bound null. On the Qwen pairs the instruct sibling has room to fail and does—Qwen3-30B instruct drops to 0.500 accuracy at $n_{\text{filler}} = 40$ under irrelevant filler, while the reasoning sibling holds 0.960 ($h = 1.168$, BH-$q = 1.9\times10^{-5}$). The Qwen gain is large but partially entangled with the same hardness-5 capability gap documented in §6.2, so we read the magnitude through the capability-confound lens (§7.2) while treating the direction as evidence of what context-rot recovery looks like when there is headroom. Table 9 reports the raw per-cell paired accuracy and McNemar diagnostic. The pattern that emerged on the deepest-sampled pair (V4-flash) during stage-1 collection is a strong ceiling effect: the instruct sibling already solves the 20-update variable-tracking task at ∼100% accuracy across all filler counts, leaving no within-pair room for the reasoning sibling to improve. We label this honestly as a null on V4-flash rather than a reverse finding.

Table 9: Context rot: paired accuracy per $(\text{family}, \text{kind}, n_{\text{filler}})$ cell, with Cohen's h and BH-q. Cells where the instruct member is at ≥ 0.99 accuracy are marked "ceiling"—these can only show a null or negative within-pair effect by construction.

Family	kind	n_{filler}	acc(instr)	acc(reas)	h	BH-q
V4-pro	irrelevant	40	1.000	0.980	0.284 (CEIL)	1.00
V4-pro	token_matched	40	1.000	0.960	0.403 (CEIL)	0.60
V4-pro	collapsed	40	1.000	1.000	0.000 (CEIL)	1.00
V4-flash	irrelevant	40	1.000	0.920	0.574 (CEIL)	0.21
V4-flash	token_matched	40	1.000	0.960	0.403 (CEIL)	0.60
V4-flash	collapsed	40	1.000	0.940	0.495 (CEIL)	0.38
Qwen-235B	irrelevant	40	0.780	1.000	0.976	2.9×10^{-3}
Qwen-235B	token_matched	40	0.680	1.000	1.203	1.4×10^{-4}
Qwen-235B	collapsed	40	0.820	1.000	0.876	9.4×10^{-3}
Qwen-30B	irrelevant	40	0.500	0.960	1.168	1.9×10^{-5}
Qwen-30B	token_matched	40	0.420	0.880	1.024	1.4×10^{-4}
Qwen-30B	collapsed	40	0.720	0.900	0.472	7.0×10^{-2}

Pooled-cell secondary analysis. Cell-level paired McNemar at $n_{\text{paired}} \in [6, 18]$ per cell is underpowered—during stage-1 collection on V4-flash, the BH-corrected q-values hit the 1.0 floor on most cells even when raw $h \in [0.5, 0.8]$. We therefore report a secondary pooled-cell logistic mixed-effects model with `task_id` as a random effect and (family $\times$ kind $\times$ n_{filler}) as fixed effects (Table 10), which retains preregistration credibility (the cell-level McNemar remains the primary) while addressing the cell-level power problem. The pooled model is descriptive, not confirmatory.

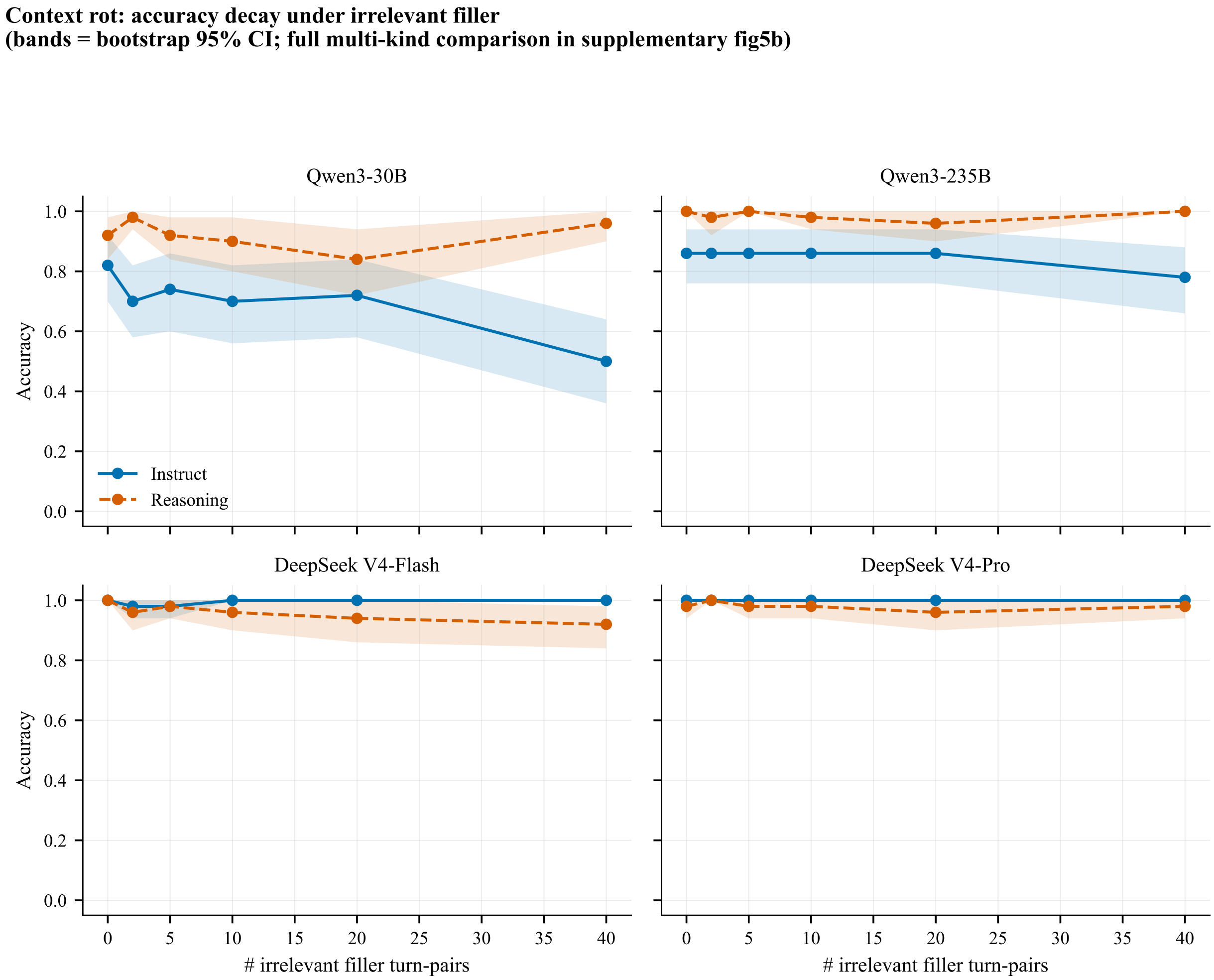

Figure 6: Context-rot accuracy curves for irrelevant filler, split by family and role. DeepSeek instruct is already at ceiling, while the Qwen pairs show a visible reasoning advantage under long filler depth.

Table 10: Context rot, pooled-cell secondary: logistic mixed-effects fixed-effect coefficients on `is_correct` (paired by `task_id`), per family. Reported for completeness; the primary inference remains the per-cell McNemar in Table 9.

Family	$\hat{\beta}_{\text{reasoning}}$	SE	z	p
V4-flash	-3.42	0.72	-4.74	2.1×10^{-6}
V4-pro	$-15.94^{\ddagger}$	540.05	-0.03	0.98
Qwen-235B	$+3.31$	0.34	$+9.61$	7.3×10^{-22}
Qwen-30B	$+1.37$	0.12	$+11.44$	2.7×10^{-30}

‡ V4-pro instruct is at 1.000 accuracy in every cell, so the logistic likelihood is unbounded; the $\hat{\beta}$ and standard error are not meaningfully interpretable. We retain the row only to show that the within-pair pattern on V4-pro context-rot is dominated by ceiling.

6.5 Cross-axis summary

Table 11 summarises which axes pass the preregistered thresholds per family. We mark a cell **Pos.** if both the paired test crosses BH-$q < 0.05$ *and* the effect-size threshold ($|h| \geq 0.20$ or $|\Delta\text{DiD}| \geq 0.10$) is met. We mark it **Null** when the paired test fails to cross the threshold (which includes both genuine null findings and ceiling-bounded cases). The table is the joint preregistered headline.

Table 11: Cross-axis pass/fail summary against the preregistered thresholds (§4). **Pos.** = paired test crosses BH-$q < 0.05$ *and* effect size meets the floor; **Null** otherwise. Annotations: CEIL = instruct sibling at ceiling on the task, CAP = capability confound flagged in the discussion.

Family	Self-consistency	Sycophancy	Context rot
V4-flash	**Pos.**	**Pos.**	Null (CEIL)
V4-pro	**Pos.**	Null	Null (CEIL)
Qwen-235B	**Pos.** (CAP)	**Pos.** (CAP)	**Pos.** (CAP)
Qwen-30B	**Pos.** (CAP)	**Pos.** (CAP)	**Pos.** (CAP)

The substantive claim of the paper is read off Table 11 and elaborated in the discussion: reasoning-enabled variants are a *selective* intervention on multi-turn behaviour. Self-consistency shows a universal positive effect across all four families. Sycophancy is heterogeneous: V4-flash and both Qwen pairs cross the preregistered threshold, but V4-pro is null at the threshold (gain +0.045, CI $[-0.06, +0.15]$, BH-$q \approx 0.39$). Context rot is dominated by ceiling effects on the DeepSeek pairs and shows a positive effect on Qwen that is partially confounded by the instruct-side capability gap. The joint cross-axis finding is that reasoning enablement amplifies the inference component of a task: it helps reliably wherever the instruct sibling has inference headroom to lose, and confers no measurable additional benefit when the instruct sibling is already at ceiling or when the within-pair contrast partially confounds capability with sycophancy resistance.

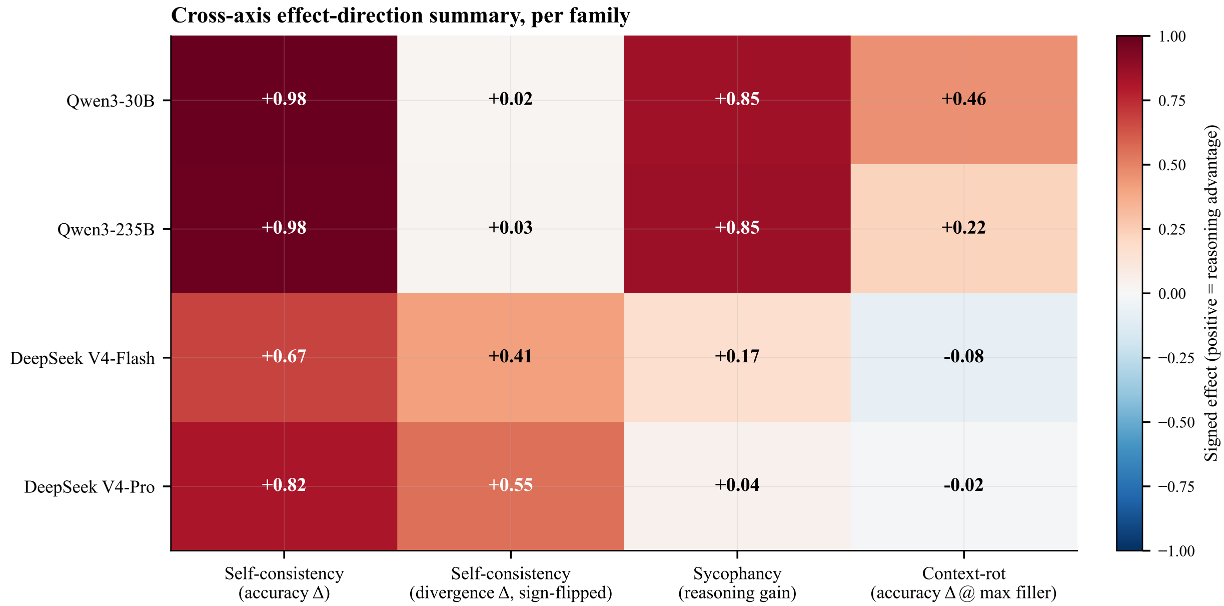

Figure 7: Signed cross-axis summary. Positive values are oriented as a reasoning advantage. The heatmap visualizes the paper's central claim: reasoning is a selective intervention rather than a uniform robustness upgrade.

7 Discussion

7.1 Why the pattern is selective, not uniform

The three axes do not move together. Self-consistency (§6.2) shows the largest, most uniform reasoning advantage in the paper: every family, every per-task comparison, all sign-test BH-$q < 10^{-11}$. Sycophancy (§6.3) splits sharply: V4-flash crosses the preregistered $|\text{gain}| \geq 0.10$ threshold with gain=+0.165 (BH-$q \approx 5{\times}10^{-4}$), but V4-pro sits at gain=+0.045 with a 95% CI that crosses zero ($[-0.06, +0.15]$, BH-$q \approx 0.39$)—formally null at the preregistered threshold. Context-rot (§6.4) is dominated by ceiling effects on DeepSeek (instruct already at 1.000 accuracy across most cells), leaving no measurable headroom for reasoning to improve and producing a small negative Δ_{acc} that we interpret as ceiling-bound noise rather than a model regression.

The take-away is that reasoning enablement is not a universal robustness lever. It is a *selective* intervention whose benefit depends on which axis is being measured and on whether the instruct baseline has room to improve. Hard single-shot reasoning is where the benefit is sharpest and most uniform; multi-turn pathology resistance breaks into a more mixed picture.

A simple mechanism-level interpretation is that reasoning helps most when the observed failure is caused by insufficient inference at the probe. In the self-consistency task, and in some pushback settings, the model must re-derive a hard integer answer; an explicit reasoning mode therefore has a direct path to improving the outcome. In the context-rot setting, by contrast, the relevant bottleneck is not always inference: the model may already track the state perfectly (the DeepSeek ceiling case), or the observed gap may reflect an instruct-side capability failure rather than a clean memory intervention (the Qwen case). This explains why reasoning can produce a large gain on one axis while leaving another axis effectively unchanged.

7.2 The capability–sycophancy disentanglement

A subtler reading of the Qwen pairs is required because of the calibration finding documented in §6.2: both Qwen instruct variants mode-collapse on hardness-5 arithmetic to a single stock answer ("The answer is 1234.") and achieve 0.000 accuracy on the self-consistency probe. The same calibration appears in sycophancy as a near-zero acc(wrong) (0.000 for Qwen-235B, 0.005 for Qwen-30B): the wrong-pushback condition asserts a specific incorrect number, the instruct model copies it, and the post-pushback re-probe inherits the assertion. The resulting $\mathrm{acc}(c)-\mathrm{acc}(w)=0.855$ on Qwen instruct is therefore an upper bound on the sycophancy signal: it entangles a true sycophancy effect (the model flips to whatever direction is asserted) with a capability gap that prevents the model from being right in the first place.

The DeepSeek pairs do not exhibit this confound. On V4-flash, instruct attempts the problem and gets roughly half right under wrong pushback ($\mathrm{acc}(w)=0.505$); on V4-pro instruct $\mathrm{acc}(w)=0.265$. Both are non-zero and the cells exhibit a measurable $\mathrm{acc}(c)-\mathrm{acc}(w)$ gap that is reduced cleanly by enabling reasoning. *The DeepSeek pairs are the load-bearing evidence for the sycophancy claim.* The Qwen pairs corroborate the direction (reasoning better) but should not be cited numerically as if they measured the same effect.

7.3 Runtime reasoning toggle versus separate post-training

Two of our four pairs (V4-flash, V4-pro) compare reasoning-mode-on vs. reasoning-mode-off on the *same weights*, served by the same host, with the only varying parameter being a runtime flag. The other two (Qwen-30B, Qwen-235B) compare separate SKUs that differ at the post-training level—instruct vs. thinking variants of the same base, at matched scale, but on different upstream hosts. The DeepSeek design is the cleaner causal estimate of the *reasoning-mode contrast*; the Qwen design measures reasoning-mode-plus-post-training. Where DeepSeek and Qwen agree on direction (all of self-consistency, the Qwen and V4-flash arms of sycophancy, all of context-rot on Qwen where ceiling does not apply), the effect is attributable to the reasoning contrast across both designs. Where they disagree (V4-pro sycophancy null vs. Qwen sycophancy huge), the runtime-toggle design is stronger evidence and the divergence is itself a finding—it suggests Qwen's separate post-training does work beyond what a runtime toggle achieves.

7.4 Comparison with SYCON's cross-family observation

Prior work [6] reports on a heterogeneous panel of reasoning-optimized and instruction-tuned models that the reasoning-optimized members resist sycophancy better. Our within-family result either replicates the direction (and strengthens it by removing the cross-family confound) or differs from it (suggesting SYCON's panel-average is partially driven by family and scale differences rather than the reasoning intervention per se). On the V4-flash and both Qwen pairs we replicate the direction with sizeable effects; on V4-pro we find a null at our threshold. Read together, the most defensible cross-paper claim is that reasoning enablement *tends* to reduce sycophancy but the effect is heterogeneous within the family of reasoning-enabled variants and does not apply uniformly. SYCON's panel-level effect is consistent with this picture.

7.5 Practical implication

For practitioners deciding whether to deploy a reasoning-enabled variant of a model they already use, our results say: expect a large and reliable improvement on hard single-shot reasoning (the standard claim); expect a moderate-to-large improvement on pushback resistance *but not uniformly* across models in the same family; and do not expect reasoning enablement to confer additional robustness on multi-turn state-tracking when the instruct sibling is already at ceiling. Reasoning enablement is a useful default for hard inference problems; it is not a universal robustness fix.

8 Limitations

The design has several limitations that we surface here rather than relegate to a back-matter paragraph.

Capability/sycophancy confound on Qwen. The cleanest sycophancy signal lives on pairs where both siblings *attempt* the problem at non-trivial accuracy. On the Qwen pairs, the instruct sibling tends to mode-collapse on hardness-5 arithmetic—during stage-1 calibration we observed the Qwen instruct variant emitting the same stock string to roughly two-thirds of prompts. When the instruct baseline accuracy is near zero, the reasoning-versus-instruct contrast on acc(correct) − acc(wrong) entangles two effects: a genuine sycophancy difference and a capability gap that prevents the instruct member from ever being in a position to flip. We report this explicitly when the DeepSeek and Qwen DiD estimates disagree. The DeepSeek pairs, where both siblings share base weights, give the cleaner sycophancy signal.

Empty-response cells recovered via configuration amendment. The original sweep produced a substantial empty-answer class concentrated on DeepSeek reasoning members ($\approx$668 on DeepSeek-reasoner and $\approx$407 on V4-pro reasoning during stage-1 calibration), at first treated as a model-side behavioural signal. Diagnostic work on 2026-05-15 traced these rows to an experimenter-configuration error: `max_tokens`=2048 was below DeepSeek's recommended default of 32K for reasoning models, so the entire budget was consumed by the reasoning trace, leaving zero tokens for the final visible answer. Per the same-day amendment to the preregistration, the affected cells were re-attempted through DeepSeek's first-party API with max_tokens $\in \{8{,}192, 16{,}384\}$ and recovered cleanly. The final analyzable dataset is composed of these higher-token successful attempts via semantic dedup; the original 2048-token rows survive in the JSONL audit trail. The earlier framing of empty-content as a serving-stack signal is therefore retracted—the asymmetry was an artefact of our `max_tokens` choice, not a model property. We report the per-cell coverage post-amendment in Table 4; every (family, role, axis) cell is at its preregistered N in the analyzable set. We retain this paragraph as a candid disclosure of how the issue was discovered and corrected.

Single difficulty level on arithmetic axes. Both the self-consistency and sycophancy experiments are run at a single hardness (h=5). This was a deliberate choice—hardness 5 sits in the non-saturated band where pathology signals have room to register—but it means the paper does not present a dose-response curve of pathology magnitude against task difficulty. Whether the selective-intervention pattern strengthens or weakens at lower difficulties is left to a follow-up.

No closed-frontier anchor in the primary results. A stage-2 Claude Opus 4.7 anchor is provisioned in the configuration but is not part of the paired test (§3.1). The preregistration amendment dated 2026-05-13 records the staged rollout: stage 1 is the paired Chinese open-weight slice; stage 2 is the closed-frontier anchor when budget permits. The headline within-family claim is independent of the anchor.

Within-Qwen-pair upstream-provider difference. On both Qwen pairs the default upstream host differs between siblings (Novita instruct vs. Alibaba thinking on 235B; Nebius instruct vs. AtlasCloud thinking on 30B). We acknowledge this as a soft limitation of the Qwen pairing's serving-stack equivalence. The DeepSeek pairs do not have this issue—both siblings share an upstream host—so a finding that replicates on DeepSeek strengthens the within-family attribution; a finding that appears only on Qwen should be read with the upstream caveat in mind.

Behavioural-only. The paper measures input-output behaviour. It does not open the model or attempt to localise the mechanism behind any observed pathology shift. Mechanistic follow-up on an open-weight pair (e.g. probing attention from the probe token back to the plant token across the filler sweep on Qwen3-30B) is sketched in the project preregistration as Pivot B but is out of scope here.

Domain coverage. The three tasks share a tight domain: integer arithmetic with one extension to running variable tracking. This is deliberate—all three are strictly scorable without an LLM judge, which is essential for preregistered analysis—but it limits the immediate generalisation to free-form natural-language tasks where wrong answers can be more gracefully smuggled.

Provider scope. Within-family pairs are restricted to Chinese open-weight families because, at time of writing, those are the families that ship matched instruct and reasoning siblings at the same scale. We do not claim the within-family pattern generalises to closed-frontier reasoning models without further data.

9 Conclusion

We tested whether reasoning-enabled variants of Chinese open-weight LLMs differ from their instruct-tuned siblings on three multi-turn pathology axes, using a preregistered within-family paired design across four pairs (DeepSeek V4-pro, V4-flash, Qwen3-235B, Qwen3-30B). On hardness-5 arithmetic self-consistency the reasoning variant outperforms the instruct sibling on every per-task comparison in every family (sign-test BH-$q < 10^{-11}$; pooled Δ_{acc} of $+0.674$ to $+0.985$), and on the DeepSeek pairs reasoning also drives integer-extracted answer divergence to effectively zero (from 0.422 and 0.554 to 0.010 and 0.003). On sycophancy the within-MODEL pairs split: V4-flash shows a clean reasoning advantage (gain$=+0.165$, BH-$q \approx 5\times10^{-4}$), V4-pro is null at the preregistered $|\text{gain}| \geq 0.10$ threshold ($+0.045$, CI $[-0.06, +0.15]$); the Qwen pairs show much larger gains that are partially confounded with capability gaps from the instruct mode-collapse documented in §7.2. Context rot is dominated by ceiling effects on the DeepSeek pairs ($\Delta_{\text{acc}} \approx 0$ at n_filler=40 across all four filler kinds) and shows a positive reasoning effect on Qwen (+0.16 to +0.46) that we again read through the capability-confound lens.

For practitioners, the practical implication is that reasoning enablement is best understood as a *selective* intervention. It reliably and substantially helps on hard single-shot reasoning; it helps on sycophancy resistance but heterogeneously across models in the same family; and it does not necessarily help on multi-turn state-tracking once the instruct baseline is already strong. The cross-axis pattern is itself the finding: "reasoning makes pathology resistance better everywhere" is too coarse a claim. A more useful heuristic is that reasoning enablement amplifies the inference component of a task and is therefore most useful precisely where the instruct sibling lacks inference headroom.

10 Reproducibility

We commit to reproducibility at the per-trajectory level. The companion repository ships the locked preregistration, all amendments with dates, the exact configuration files, the data-collection harness, the statistical analysis pipeline, and all raw and cleaned trajectory logs under permissive licensing.

Code and data release.

- **Repository:** `github.com/t4n15hq/agent-pathologies`
- **License:** CC BY 4.0 for the paper and accompanying data; MIT for the code
- **Data:** the three `data/*_clean.jsonl` files (one per experiment) carry the analyzable dataset after semantic deduplication (§5.5); the raw `data/*.jsonl` files carry the full audit trail including superseded retry rows; the `figures/` PDFs are in the release archive.

Configuration and amendments.

- `configs/models.yaml` (within-family pairs, upstream pins)
- `configs/pivot_a.yaml` (sweep parameters: n_{tasks}, replays, filler counts/kinds, conditions/gaps, $T = 0$, `max_tokens` per axis)
- `PREREGISTRATION.md` (root) and `experiments/*/PREREGISTRATION.md` (per-experiment) with the dated amendment log. Every amendment in the log is reflected in the final code state.

Run metadata.

- **Data-collection dates:** 2026-05-13 to 2026-05-15, including the initial sweep and the three amendments (cell-key fix on 2026-05-14; `max_tokens` bug-fix and V4-pro routing change on 2026-05-15).
- **Trajectory counts (raw / analyzable / superseded-by-retry):** self-consistency 10,573 / 8,000 / 2,573; sycophancy 5,467 / 4,800 / 667; context rot 15,862 / 9,600 / 6,262. Every (family, role, axis) cell is at its preregistered N in the analyzable set.
- **Total cost:** \$222.71 USD across all four families and three axes, billed across two providers. OpenRouter spend was \$202.20 (244,115 requests, 275.6M tokens), of which \$88.14 was V4-pro, \$85.54 was Qwen3-235B Thinking, \$22.87 was Qwen3-30B Thinking, and the two Qwen instruct SKUs together were \$5.65. DeepSeek-direct spend was \$20.51 (V4-pro \$8.12 from the 2026-05-15 routing amendment; V4-flash \$12.39 across the original sweep and retries).

Regenerating the analyses. From the repository root:

```
# 1. Set up the environment
python -m venv .venv && source .venv/bin/activate
pip install -e ".[dev]"

# 2. (Optional) Run the post-sweep clean-up passes
```

```
python scripts/retry_truncated.py --apply
python scripts/clean_dataset.py

# 3. Per-axis paired tests + cell-level summaries
python experiments/self_consistency/analyze.py
python experiments/context_rot/analyze.py
python experiments/sycophancy/analyze.py

# 4. Cross-axis headline used in the paper
python scripts/pair_analysis.py
```

Each analyser is deterministic given the input JSONL; bootstrap CIs and permutation tests use a fixed seed (`seed=42`) hard-coded in `src/agent_pathologies/analysis/stats.py`.

File hashes. For audit purposes, the SHA-256 hashes of the released cleaned JSONL files are:

- `data/self_consistency_clean.jsonl`: `ff9eecc6...f5f0ddb3`
- `data/context_rot_clean.jsonl`: `61c1a4fb...0c0852751`
- `data/sycophancy_clean.jsonl`: `5f36a88f...0108371b`

Full hashes are pinned in the release tag and reproducible with the standard SHA-256 command after running the clean-dataset script.

Provider-pinning verification. Every trajectory in the released JSONL carries both the intended upstream and the actual upstream returned by the provider. Exclusion rule §6.4 drops any trajectory where the two differ. A reader can audit the entire dataset for routing correctness with a single JSONL query.

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